

Le Nhu Dat

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SUMMARY

AI Engineer focused on RAG and LLM agents that help businesses turn complex documents into actionable answers and automate knowledge-intensive workflows. Brings hands-on experience building and deploying a RAG application with answer verification and quality evaluation, and is extending that foundation into agents for multi-step tasks.

EXPERIENCE

AI Engineer Intern

Ho Chi Minh City, Vietnam

TMA Solutions – Vietnamese software services company

March 2026 – May 2026

- Developed an end-to-end sign language recognition pipeline to convert sentence-level signing videos into word sequences, covering data preprocessing, CorrNet modeling with MediaPipe keypoints, training, and evaluation
- Improved preprocessing for variable-length sequences and missing keypoints, increasing word-level accuracy within sentences from 90% to 94% and reducing word error rate (WER) from 9% to 7% on How2Sign
- Adapted the pipeline to the company's sign language dataset, retraining the model from scratch and tuning its architecture and hyperparameters

PROJECTS

Vietnamese Legal QA – Production RAG

github.com/lndat18/production-legal-qa-rag

- Independently built and deployed a Vietnamese legal Q&A chatbot from document ingestion through evaluation and serving, covering six laws and 2,228 structure-aware chunks with Article/Clause citations
- Combined dense search and BM25 with HyDE, Reciprocal Rank Fusion, and a local Vietnamese reranker to retrieve semantic matches and exact legal references
- Added citation and number checks plus an independent LLM evidence judge to check answers against retrieved evidence; allowed one repair attempt and refused answers that failed verification
- Evaluated retrieval and generation on 157 synthetic questions: Context Recall 0.86 and Faithfulness 0.83 on 143 answered cases; 14 cases were refused. Built checkpointed evaluation runs with API-key rotation and quota-aware throttling to resume across provider limits
- Delivered an OpenAI-compatible FastAPI API and OpenWebUI interface with Docker Compose and Cloudflare Tunnel; added Redis caching, request limits, backups, tracing, and metrics, with GitHub Actions CI running 1,700+ tests

Agentic GraphRAG for Legal Q&A — In Progress

github.com/lndat18/production-legal-agentic-graph-rag

- Extending the first project's retrieval and task capabilities, reusing its structured legal corpus, hybrid search, answer verification, conversation handling, caching, API, deployment, and monitoring
- Built a Neo4j knowledge graph linking Articles, Clauses, and Points through legal cross-references; currently testing graph retrieval to gather connected provisions beyond similarity search
- Developing more flexible retrieval with LangGraph: a planner decomposes complex questions, while hybrid and graph retrievers gather evidence and follow cross-references across Clauses, Articles, and documents
- Designing tool-based workflows that extend the chatbot from answering questions to completing tasks, starting with personal income tax calculation and DOCX/PDF form drafts, using deterministic arithmetic, legal citations, and effective-date checks
- Planning Jev for routing decisions and MCP for tool integration, with RAGAS/DeepEval evaluation against the original RAG to measure evidence coverage, tool-call accuracy, task completion, numeric correctness, latency, and cost

EDUCATION

TON DUC THANG UNIVERSITY

Ho Chi Minh City, Vietnam

Bachelor of Computer Science

2022 – 2026

GPA: 3.06/4.00 Classification: Good

TECHNICAL SKILLS

Programming Languages: Python, C, Java

ML & Deep Learning Frameworks: TensorFlow/Keras, PyTorch

LLM Development Frameworks: LangChain, LangGraph

Agent Routing & Tool Integration: Jev, MCP

Model Platforms & Inference APIs: Hugging Face, Groq

Databases & Data Stores: Neo4j, PostgreSQL, Pinecone, Redis

Model Deployment & Serving: FastAPI, Docker, Cloudflare Tunnel, GitHub Actions (CI/CD)

Monitoring & Evaluation: Langfuse, Prometheus, Grafana, RAGAS, DeepEval

CERTIFICATIONS

- [IBM RAG and Agentic AI](#) — IBM
- [IBM Generative AI Engineering Professional Certificate](#) — IBM
- [LLMOps \(Large Language Model Operations\)](#) — Duke University
- [MLOps \(Machine Learning Operations\)](#) — Duke University
- [AIO2025](#)
- [Retrieval Augmented Generation \(RAG\)](#) — DeepLearning.AI
- [Natural Language Processing Specialization](#) — DeepLearning.AI
- [Deep Learning Specialization](#) — DeepLearning.AI
- [Machine Learning Specialization](#) — DeepLearning.AI
- [Introduction to Git and GitHub](#) — Google
- [AWS Certified Cloud Practitioner](#) — Amazon Web Services — In Progress

ADDITIONAL INFORMATION

English: [IELTS 5.5 \(CEFR B2\)](#)